When Should Momentum Be Reset?

Adaptive Restart and Hamiltonian Optimization – STAT 605 Project Proposal

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October 29, 2026

Accelerated Gradient Descent

- In the previous lecture, we learned about Nesterov's momentum.

$$x_{k+1} = y_k - \eta \nabla f(y_k) \tag{1}$$

$$y_{k+1} = x_{k+1} + \beta_{k+1}(x_{k+1} - x_k) \tag{2}$$

- If we choose optimal β_k, then the AGD with $\eta = 1/L$ can achieve

	L-smooth & convex	L-smooth & μ-strongly convex
GD	$O(1/k)$	$O((1-1/\kappa)^k)$
AGD	$O(1/k^2)$	$O((1-1/\sqrt{\kappa})^k)$

for the objective gap $f(x_k) - f^\star$.

How Can We Choose β_t?

Algorithm 1

Initialize $x_0 = y_0$ and $\theta_0 = 1$.

for $k = 0, 1, \ldots$ **do**

$$x_{k+1} = y_k - \eta \nabla f(y_k)$$
$$\theta_{k+1} = (1 + \sqrt{1 + 4\theta_k^2})/2$$
$$\beta_{k+1} = (\theta_k - 1)/\theta_{k+1}$$
$$y_{k+1} = x_{k+1} + \beta_{k+1}(x_{k+1} - x_k)$$

end for

Algorithm 2

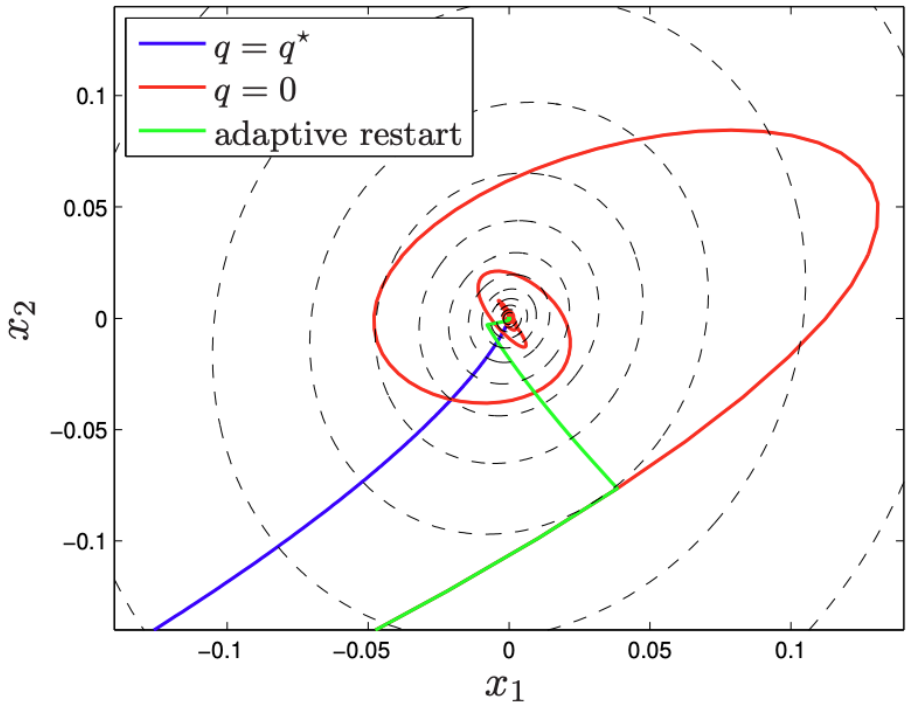
Initialize $x_0 = y_0$, $\theta_0 = 1$ and $q \in [0, 1]$.

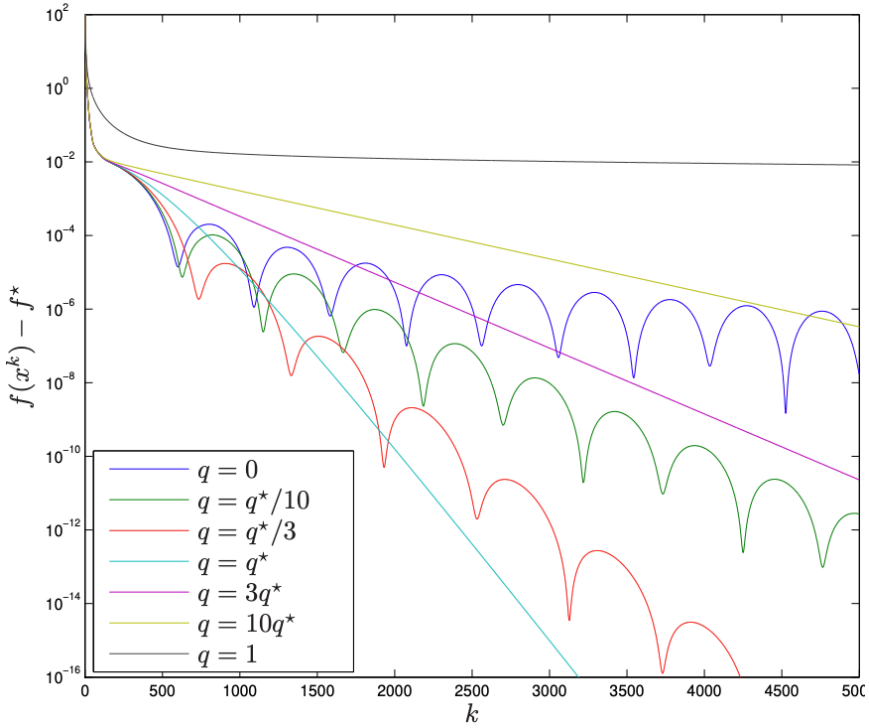
for $k = 0, 1, \ldots$ **do**

$$x_{k+1} = y_k - \eta \nabla f(y_k)$$
$$\theta_{k+1} = (1 - q\theta_k^2 + \sqrt{(1 - q\theta_k^2)^2 + 4\theta_k^2})/2$$
$$\beta_{k+1} = (\theta_k - 1)/(\theta_{k+1} + q\theta_k^2)$$
$$y_{k+1} = x_{k+1} + \beta_{k+1}(x_{k+1} - x_k)$$

end for

- The optimal choice for q is
  - 0 for L-smooth and convex function.
  - μ/L for L-smooth and μ-strongly convex function.
- In practice, we often assume that μ is zero and use Algorithm 1. However, this can dramatically slow down the convergence of the algorithm.


$q = q^\star$
$q = 0$
adaptive restart
0.1
0.05
0
−0.05
−0.1
−0.1
−0.05
0
0.05
0.1
x_2
x_1


10^2
10^0
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
$f(x^k) - f^\star$
0
500
1000
1500
2000
2500
3000
3500
4000
4500
k
$q = 0$
$q = q^\star/10$
$q = q^\star/3$
$q = q^\star$
$q = 3q^\star$
$q = 10q^\star$
$q = 1$

Restart the Momentum

- Usually we don't know μ.
- An alternative to choosing the optimal value of q is to use restarting.
- For example, we can restart the algorithm every k iterations.
- O'donoghue and Candes (2015) suggested two adaptive restart methods
  - Restart whenever
$$f(x_k) > f(x_{k-1}). \tag{3}$$
  - Restart whenever
$$\nabla f(y_{k-1})^\top (x_k - x_{k-1}) > 0. \tag{4}$$

Motivation: Optimization as Particle Motion

Consider the unconstrained problem

$$\min_{x \in \mathbb{R}^n} f(x), \qquad f \text{ smooth and convex.}$$

A mechanical interpretation of momentum methods is

$$\underbrace{\ddot{x}}_{\text{acceleration}} = \underbrace{-\nabla f(x)}_{\text{force}} - \underbrace{B(t)\dot{x}}_{\text{viscous damping}}, \qquad B(t) \succeq 0.$$

- The objective f acts as potential energy.
- Inertia allows the particle to accumulate velocity.
- Friction dissipates energy and controls oscillations, but also slows the motion.

The paper's proposal: allow frictionless motion and remove momentum at carefully chosen restart times.

Physical mechanism: Motion without Friction

A particle of unit mass starts at rest:

$$\ddot{x} + \nabla f(x) = 0, \qquad x(0) = x_0, \quad \dot{x}(0) = 0.$$

Its total energy is conserved:

$$H(t) = f(x(t)) + E_K(t) = f(x_0), \qquad E_K(t) = \frac{1}{2}\|\dot{x}(t)\|^2.$$

Consequently,

$$\boxed{f(x_0) - f(x(t)) = E_K(t).}$$

- Objective reduction becomes kinetic energy.
- Frictionless motion alone can produce persistent oscillations.
- A restart keeps the position and sets $\dot{x} = 0$, removing the accumulated kinetic energy.

Piecewise conservative: energy is conserved between restarts and lost at restarts.

Restart Criterion: Average Progress

A natural idea is to restart at a maximum of kinetic energy. However, controlling the waiting time is difficult in several dimensions.

The paper instead measures objective reduction per unit time:

$$\boxed{r(t) = \frac{E_K(t)}{t} = \frac{f(x_0) - f(x(t))}{t}.}$$

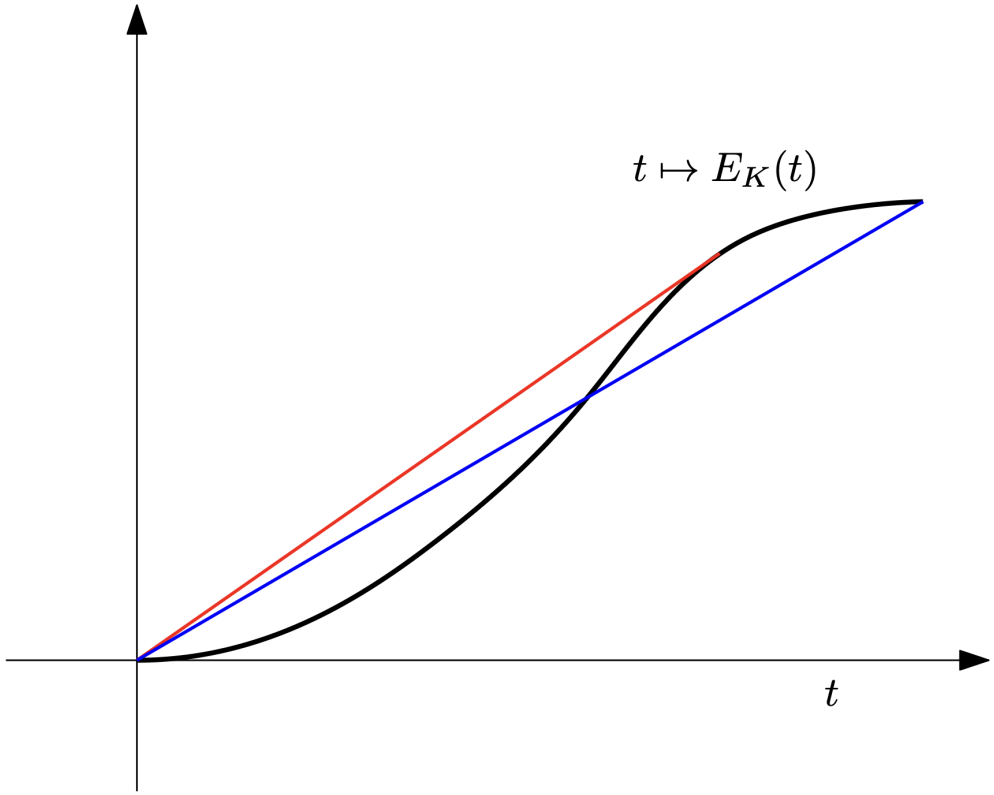
Restart at the first local maximum of r:

$$r'(t) = \frac{tE_K'(t) - E_K(t)}{t^2}.$$

- Continue while the average rate of improvement grows.
- Restart when that rate begins to decrease.
- Measure elapsed time from the latest restart.

Key idea: balance the amount of progress against the time spent obtaining it.

Maximum Mean Dissipation



$$\frac{E_K(t)}{t} = \text{slope of the line from the origin to } (t, E_K(t)).$$

The steepest line identifies the greatest average improvement. In this illustration, it occurs before the kinetic-energy peak.

Source: A. Scagliotti and P. Colli Franzone, *A Piecewise Conservative Method for Unconstrained Convex*

Discrete algorithm and main results

Symplectic Euler produces a gradient step with momentum:

$$v_{k+1} = v_k - h\nabla f(x_k), \qquad x_{k+1} = x_k + hv_{k+1}.$$

Between restarts, this is equivalent to

$$x_{k+1} = x_k - h^2\nabla f(x_k) + (x_k - x_{k-1}).$$

Continuous-time convergence. For $f \in C^2$, L-smooth and μ-strongly convex, maximum mean dissipation restarts yield

$$\boxed{f(\widetilde{x}(t)) - f^\star = O(e^{-ct}), \qquad c = \frac{\mu^{3/2}}{32L}\log\left(1 + \frac{\mu}{L}\right).}$$

A sufficient evolution time to reduce the initial error by a factor $\varepsilon \in (0, 1)$ is

$$t = O\left(\frac{L^2}{\mu^{5/2}}\log\frac{1}{\varepsilon}\right).$$

The rate depends on μ, but the restart rules do not require knowing μ.

From Sampling to Optimization: Hamiltonian Flow

- Hamiltonian Flow (HF) inspired Hamiltonian Monte Carlo (HMC) for sampling. Can we go the other way and use HF ideas to design **optimization** algorithms?
- Hamiltonian dynamics for $\min_{x \in \mathbb{R}^d} f(x)$:

$$\dot{X}_t = Y_t, \qquad \dot{Y}_t = -\nabla f(X_t)$$

  where Y_t is the velocity and the gradient drives the acceleration.
- The total energy $H(X_t, Y_t) = f(X_t) + \frac{1}{2}\|Y_t\|^2$ is **conserved**, so the particle oscillates forever and never settles at the minimizer.
- **Fix:** periodically reset the velocity to 0. Starting from $H(x_k, 0) = f(x_k)$,

$$f(x_k) = f(x_{k+1}) + \tfrac{1}{2}\|\tilde{y}_{k+1}\|^2 \quad \Longrightarrow \quad f(x_{k+1}) \leq f(x_k).$$

- Each reset removes kinetic energy, so f decreases. *When* should we reset?

Randomized Hamiltonian Flow (RHF)

- Reset at a random integration time $\tau_k \sim \mathrm{Exp}(\gamma(T_k))$, where T_k is the accumulated time. Larger $\gamma \Rightarrow$ more frequent resets.
- Resets form a Poisson process N_t with rate $\gamma(t)$:

$$dX_t = Y_t\,dt, \qquad dY_t = -\nabla f(X_t)\,dt - Y_{t^-}\,dN_t$$

  where $dN_t = 1$ if a reset occurs in $[t, t+dt)$ and 0 otherwise.
- This is a *randomized* version of restarting the momentum.
- **Convergence rates:**

	Reset rate $\gamma(t)$	Rate
μ-strongly convex	$\sqrt{16\mu/5}$	$O\left(e^{-\sqrt{\mu/5}\,t}\right)$
Convex	$6/(t+1)$	$O\left(1/(t+1)^2\right)$

- These match the accelerated rates of AGD.

Randomized Hamiltonian Gradient Descent (RHGD)

Approx. 1: Bernoulli resets. Over a step of length h,

$$P(\text{reset within } h) = 1 - e^{-\gamma h} \approx \gamma h,$$

so reset with probability $p_k = \min(\gamma_k h, 1)$.

Approx. 2: Discretize the flow.

$$\frac{x_{k+1} - x_k}{h} = y_{k+1}, \qquad \frac{y_{k+1} - y_k}{h} = -\nabla f(x_{k+1})$$

gives $x_{k+1} = x_{k+\frac{1}{2}} - h^2 \nabla f(x_{k+1})$ with $x_{k+\frac{1}{2}} = x_k + h y_k$. This implicit (proximal) step is hard, so use $\nabla f(x_{k+1}) \approx \nabla f(x_{k+\frac{1}{2}})$.

Algorithm 3 (RHGD)

Initialize x_0, $y_0 = 0$, step $h > 0$, rate $\gamma_k > 0$.

for $k = 0, 1, \ldots, K-1$ **do**

$$x_{k+\frac{1}{2}} = x_k + h y_k$$
$$x_{k+1} = x_{k+\frac{1}{2}} - h^2 \nabla f(x_{k+\frac{1}{2}})$$
$$\tilde{y}_{k+1} = y_k - h \nabla f(x_{k+1})$$
$$y_{k+1} = \begin{cases} 0 & \text{w.p. } \min(\gamma_k h, 1) \\ \tilde{y}_{k+1} & \text{otherwise} \end{cases}$$

end for

return x_K

Prior Work: When Should We Reset Momentum?

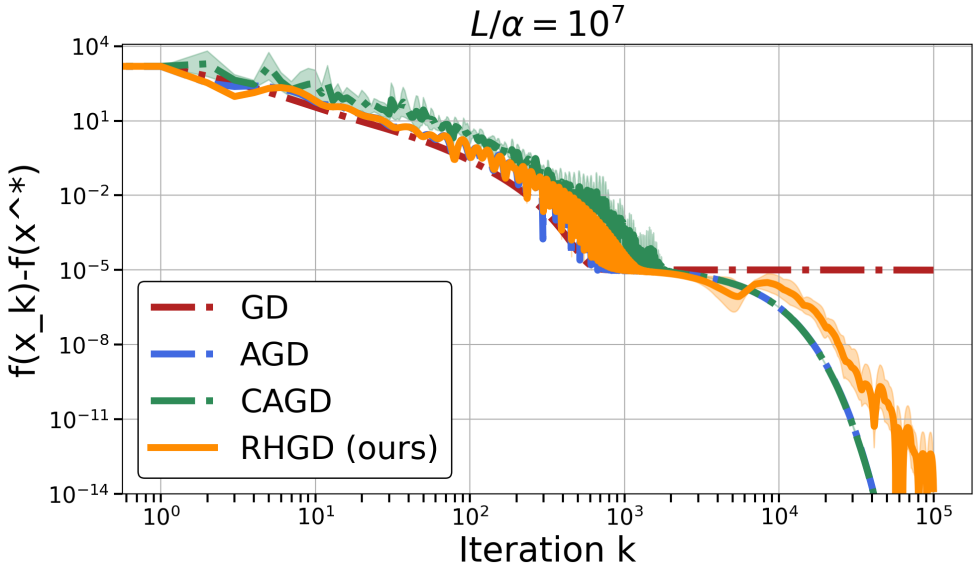
- Three approaches offer different rules for discarding momentum:
  - **Adaptive AGD:** restart using objective increase or gradient alignment.
  - **RHGD:** reset velocity randomly, with probability $p_k = \min\{\gamma_k h, 1\}$ at each step.
  - **Conservative methods:** restart using trajectory information, e.g., kinetic energy, mean dissipation, or gradient alignment.
- **The trade-off:** resetting can suppress unhelpful oscillations, but resetting too early may discard useful momentum.

Common theme

Should momentum resets follow a prescribed (possibly randomized) schedule, or adapt to information observed along the trajectory?

Motivation: Can Resetting Be Improved?

- **Theory:** RHGD achieves accelerated complexity $O(\sqrt{\kappa}\log(1/\varepsilon))$ in expectation.
- **Empirical evidence:** RHGD is competitive, but its relative performance varies across regimes. With known μ, AGD/CAGD can be faster; with misspecified μ, RHGD can be more robust.
- **Opportunity:** RHGD uses prescribed refresh probabilities rather than observed trajectory information.



Fu & Wibisono (2025), Fig. 1 (left): $\kappa = 10^7$, exact μ; tuned stepsizes.

Main research question

Can **trajectory-dependent, μ-free resets** improve RHGD's **practical efficiency and robustness**, and under what conditions?

A Possible Mechanism: Spectral Geometry

- Consider strongly convex quadratics with identical d, μ and L:

$$f(x) = \tfrac{1}{2} x^\top A x, \qquad A = Q \operatorname{diag}(\lambda_1, \ldots, \lambda_d) Q^\top, \quad \mu \le \lambda_i \le L.$$

- Starting from zero velocity, each eigendirection oscillates at its own frequency $\omega_i = \sqrt{\lambda_i}$:

$$z_i(t) = z_i(0) \cos(\omega_i t), \qquad E_K(t) = \tfrac{1}{2} \sum_i \lambda_i z_i(0)^2 \sin^2(\omega_i t).$$

- **One global reset affects all modes simultaneously.** Signals such as $E_K(t)$ or $E_K(t)/t$ combine these different frequencies.

Hypothesis to test

Even at fixed $\kappa = L/\mu$, the **eigenvalue distribution** and **initial modal energy** may influence the relative effectiveness of different reset strategies.

Compare linear, log-spaced, and clustered spectra with identical endpoints; Fu & Wibisono's quadratic examples use linear spacing.

Planned Experiments: From Replication to New Resets

1. **Replicate:** RHGD's representative quadratic results, plus selected RCM and restarted-AGD benchmarks.
2. **Compare fairly:** AGD, restarted AGD, CAGD, RHGD, and RCM methods, counting *gradient evaluations* as well as iterations.
3. **Diagnose:** vary eigenvalue distributions while holding (d, μ, L) fixed; test stepsize and μ misspecification.
4. **Test adaptive resets:** hold the RHGD position/velocity updates fixed; compare prescribed random, fixed-time, gradient-alignment, and power-normalized kinetic-energy restart rules.
5. **Extend (if time permits):** evaluate promising rules on regularized logistic regression.